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Original Article

Adaptive water use strategies of Mongolian pine plantations in response to mining subsidence across two soil types: a case study in Northwest China

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Abstract

Understanding plant water use strategies is critical for managing and restoring ecosystems affected by coal mining subsidence. However, how subsidence impacts water use strategies, especially across different soil types, remains inadequately explored. To address this gap, we employed a continuous isotopic mixing model (based on δ^2H and δ^{18}O), coupled with δ^{13}C, soil water content and root distribution, to investigate the differences in water use strategies of Mongolian pine plantations in coal mining subsidence and non-subsidence areas with sandy and loess soils. Our results show that subsidence induces preferential flow, increases deep soil water (>80 cm), and enhances root growth and soil water-root coupling, especially in the loess areas. Isotopic mixing modeling revealed that in sandy areas, deep soil water uptake was similar between non-subsidence (79.43 ± 3.83%) and subsidence (82.69 ± 1.52%) plots. In loess areas, subsidence plots (26.36 ± 1.98%) had significantly higher deep water uptake than non-subsidence plots (16.23 ± 1.91%, $P < .01$). Leaf δ^{13}C values decreased significantly in both soil types under subsidence, indicating reduced water stress via deep water utilization, particularly in loess areas. Soil-type dependent response highlights the necessity for distinct vegetation maintenance or restoration strategies in subsided areas across different soil matrices. These findings advance understanding of plant survival strategies and water resource relationships in subsidence zones, providing valuable references for sustainable land and water management in mining-impacted areas.

Keywords subsidence, soil type, deep soil, water use strategy, stable isotopes

Introduction

Coal remains a crucial energy source for many countries, playing an indispensable role in industrial and social development (Wu *et al.* 2024, IEA 2025). However, a series of environmental issues arising from underground coal mining have attracted widespread attention. Underground coal mining typically leads to the formation of large-scale underground goafs, which induce land subsidence and alter the natural environment of the Earth's Critical Zone (ECZ), particularly affecting soil hydrological processes in the vadose zone and thereby disrupting vegetation dynamics (Yang *et al.* 2023a, Davydzenka *et al.* 2024). Given the ECZ's key role in sustaining terrestrial ecosystems and supporting human survival and development (Richter and Billings 2015), understanding the impacts of mining-induced subsidence on its various components is essential.

The root-soil interface serves as a critical junction within the ECZ and should be recognized as a vital component of surface ecosystems (Tomobe *et al.* 2021, Demir *et al.* 2024). Plant roots—encompassing their spatial distribution, morphological traits, and physiological activity—act as the direct medium for plants to absorb water and nutrients (Addo-Danso *et al.* 2020, Agee *et al.* 2021). Soil water, meanwhile, functions as a key regulator in the hydrological cycle, supporting plant growth through direct water supply and thus influencing plant health and productivity (Ali *et al.* 2021, Pokhrel *et al.* 2021). The mutual feedback between these two components at the root-soil interface collectively determines plant water uptake processes, providing valuable insights into plant survival mechanisms and adaptability to environmental changes (Wu *et al.* 2022, Jia *et al.* 2024). Therefore, understanding the responses of plant roots, soil water, and plant water

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uptake patterns to mining-induced subsidence is crucial for advancing horticultural, ecological restoration, and soil management practices in mining-affected areas.

The impacts of mining-induced subsidence on the soil–plant continuum have long been a key research focus (Bai *et al.* 2022), as vegetation stabilization are vital for ecological recovery in mining-affected regions, with broader implications for social stability (Li *et al.* 2013). Current studies have presented conflicting views on how mining subsidence affects soil hydraulics. Some research indicates that subsidence accelerates surface evaporation and reduces soil water storage (Zhang *et al.* 2022), while others report no significant differences in soil moisture between subsided and non-subsided areas (Wang *et al.* 2024). Numerical simulations and experimental studies suggest that coal mining subsidence may enhance infiltration, facilitating rapid recharge of deep soil layers through preferential flow (Fawang *et al.* 2012, Stewart *et al.* 2016). Similarly, debates persist regarding the effects of subsidence on plant roots. Some argue that subsidence causes vegetation degradation, hinders recovery, and increases mortality (Liu *et al.* 2020, Du *et al.* 2021), while others contend that ecosystems possess inherent self-regulatory mechanisms enabling plants to maintain stable growth through adaptive changes in root morphology (He *et al.* 2017, Wei *et al.* 2025). Studies investigating plant water uptake sources by integrating coupled changes in soil moisture and roots have also failed to reach consistent conclusions. Some research conducted in the Mu Us Sandy Land shows that subsidence enables *Artemisia desertorum* to access deeper soil water (Chen *et al.* 2022), whereas others report no significant differences in the water uptake patterns of *Stipa bungeana* between subsided and non-subsided areas (Wei *et al.* 2024).

However, these studies have primarily focused on shallow soil layers (<1 m), which do not fully represent the vadose zone or the entire root zone with the roots of shrubs or trees typically extending beyond 1 m (Fan *et al.* 2017, Laughlin *et al.* 2023). Experimental evidence indicates that insufficient consideration of root depth can lead to biases in water source partitioning results and distort interpretations of plant water use strategies (Tao *et al.* 2023). Meanwhile, growing evidence highlights the critical role of deep soil water in plant adaptability to seasonal droughts and environmental stress (Gessler *et al.* 2022, He *et al.* 2024). Nevertheless, the impacts of coal mining subsidence on deep soil water and deep plant water uptake remain inadequately studied. Additionally, soil water-holding capacity and hydraulic conductivity vary with soil type, serving as key factors limiting plant access to soil moisture (Kodur 2017, Gu *et al.* 2025). Current research often overlooks changes in soil type, instead focusing on subsidence effects in a single soil type—hindering a comprehensive understanding of how subsidence regulates soil water, plant roots, and plant water use strategies.

Mongolian pine (*Pinus sylvestris* var. mongolica), an evergreen coniferous tree species with deep root systems and high drought resistance, is widely used in ecological restoration projects in northern China to combat desertification (Pei *et al.* 2023) and was therefore selected as the study object. We hypothesize that the response of Mongolian pine's water use strategies to subsidence varies with soil type. The Shendong Mining Area, located at the junction of the Mu Us Sandy Land and the Loess Plateau in China, features uniform climatic conditions but two distinct soil types, coupled with severe land subsidence caused by underground coal mining—making it an ideal natural laboratory for addressing the research gaps outlined above. Using hydrogen, oxygen, and carbon stable isotope techniques, we compared the effects of mining-induced land subsidence on (i) deep soil water changes, (ii) root distribution, and (iii) water use strategies of Mongolian pine stands across different soil types. The findings of this study will enhance our understanding of plant-environment interactions under subsidence conditions and lay a theoretical foundation for formulating ecological restoration and vegetation management strategies in land subsidence areas.

Materials and methods

Study area

The research was conducted in the transitional area linking the Mu Us Sandy Land and the Loess Plateau. This particular region is typified by the coexistence of wind erosion and water erosion. It functions as a vital interface connecting the loess hilly ecosystem and the sandy land ecosystem (Fig. 1a). The area is characterized by a temperate continental climate, with a distinct feature of extreme hydrological imbalance. The average annual rainfall in this area is under 400 mm, whereas the evapotranspiration surpasses 1500 mm (Pei *et al.* 2023). Simultaneously, this region is an important energy hub with high intensity coal mining activities. Two representative study sites were chosen: The Daliuta Coal Mine belongs to the Mu Us Sandy Land and the Buertai Coal Mine belongs to the Loess Hilly Region. Based on USDA classification, the soil profile in Daliuta Coal Mine is vertically homogeneous with a consistent composition of clay (3.7%–7.3%), silt (0.2%–7.7%), and sand (85.3%–92.8%) throughout its depth. In contrast, for the soil profile in Buertai Coal Mine, the shallow upper part is still dominated by sand, while the lower part has a lower sand content (35.6%–60.7%) and a higher silt content (33.4%–58.4%). Both mining areas have experienced extensive underground coal extraction processes. As a consequence, the extraction process creates underground voids that drain groundwater, creating a cone of depression. This leads to phenomena such as surface subsidence, fissure formation, and a substantial decline in the groundwater table.

In total, four experimental plots were established in the two soil zones (two per soil zone). The plots are stocked with young Mongolian pine plantations (around 6 years old) in both subsided and non-subsided areas of coal mining. Grasses, dominated by *Stipa* species, were grown as associated vegetation between the rows of Mongolian pines. No artificial structures were present within any of the experimental plots, and the specific information for each plot is provided in Table 1. The subsidence zones had stabilized with no new structural deformations recorded at present.

Sample collection and analysis

Within each plot, three sampling spots were chosen for collecting root and soil samples. For non-subsidence plots, the three sampling locations were precisely positioned at the midpoint between paired trees to ensure consistent and representative sampling across the undisturbed area. In contrast, for subsidence plots, stricter selection criteria were applied to ensure that the collected soil profiles were clearly affected by subsidence disturbances. In addition to following the midpoint criterion, the selected locations were near intersections with ground fissures (Fig. 1c and d). For plant sample collection, sampling was conducted on the paired trees corresponding to the soil sampling points. We assume that this approach enables accurate capture of environmental changes directly caused by subsidence, providing critical insights into the impact on plant water use strategies.

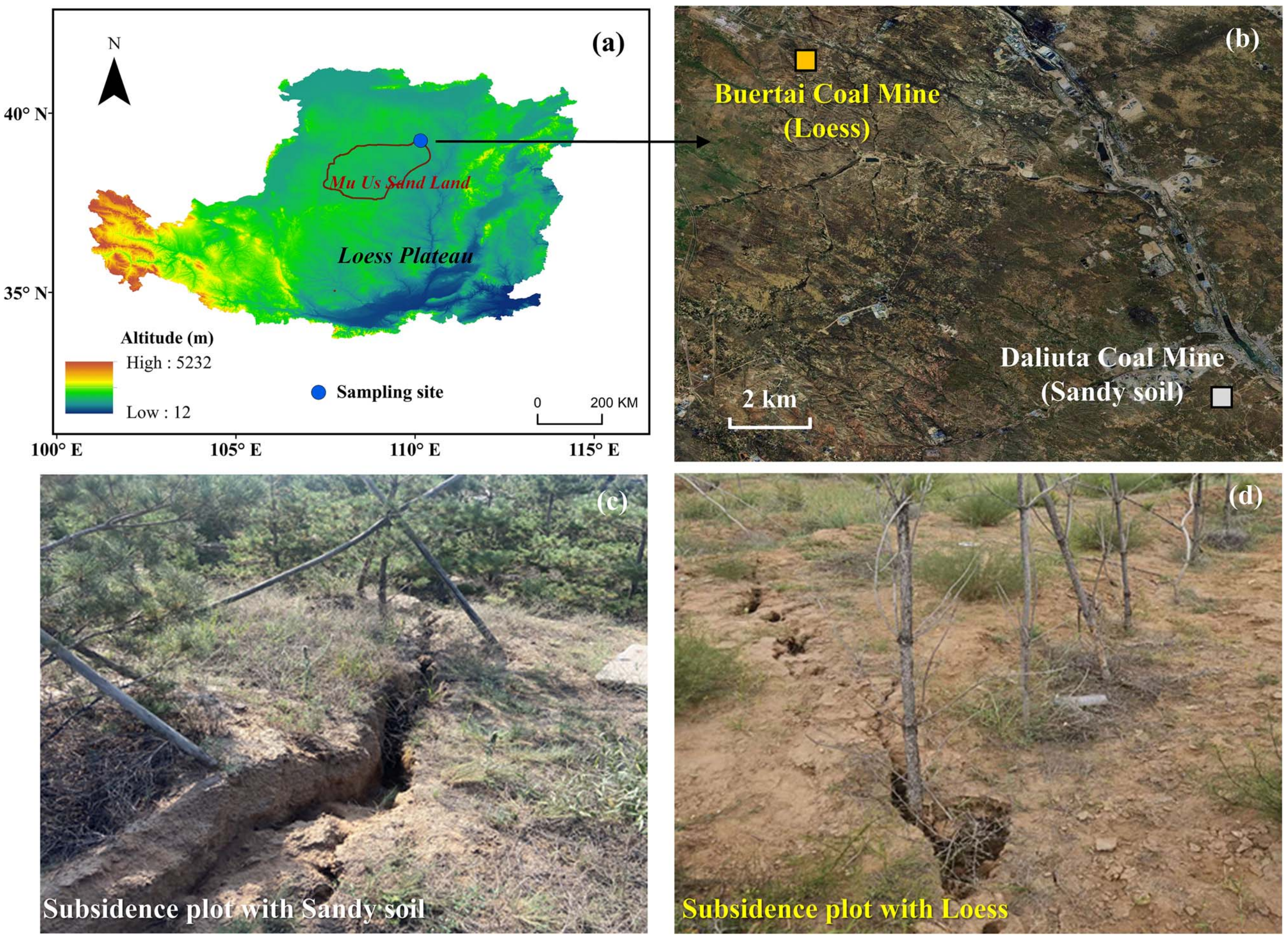

Figure 1 Location of study area (a), sampling sites of sandy soil and loess (b), and on site pictures of subsidence plot with sandy soil (c) and loess soil (d).

Table 1 Basic information of the plots.

Soil type	Plot	Plot area (m)	Plant height (m)	Crown diameter (m)	DBH (cm)	Plant density (trees ha^{-1})
Sandy soil	Non-subsidence plot	6×12	2.2 ± 0.4	0.7 ± 0.1	6.2 ± 0.3	2083
	subsidence plot	6×12	2.4 ± 0.3	0.8 ± 0.3	6.0 ± 0.5	2083
Loess soil	Non-subsidence plot	4×12	2.0 ± 0.2	0.6 ± 0.1	5.7 ± 0.5	3333
	subsidence plot	4×12	1.9 ± 0.4	0.5 ± 0.2	5.5 ± 0.4	3333

Root-soil samples were collected by using a hand auger (90 mm inner diameter) to drill to a depth of ~3 m, at which point no roots were observed. Samples were taken at 20 cm intervals throughout the drilled core. One hundred and eighty root-soil samples were gathered from 12 profiles. These samples were passed through a 1 mm sieve and rinsed with tap water to detach the roots from the soil particles. During collection, roots were inevitably cut by the auger. For the subsequent analysis, we included all root fragments retained on the sieve. Only fresh roots (judged by their plump texture and absence of rot or desiccation) were carefully picked out with tweezers, with clear distinction made between Mongolian pine roots (characterized by their black color) and grass roots (noted for their white color) based on their morphological color differences. Root fragments were considered for measurement regardless of whether they were whole or partial, as the scanning and software analysis measures the total length of all root segments present. The selected roots were scanned at a resolution of 300 dpi, and then analyzed using RhizoVision Explorer 2.0.3 (Seethepalli *et al.* 2021) with the 'Broken Roots' analysis mode to gauge the length of fine roots (Diameter < 1 mm). Fine roots are the main structures through which plants take in water (Addo-Danso *et al.* 2020). The fine root length density (FRLD) was determined by dividing the total length of fine roots by the volume of the soil specimen.

To investigate plant water use in response to coal mining subsidence under varying hydrological conditions, soil and xylem samples were collected on July 29 (S1), July 31 (S2), August 2 (S3), and August 4 (S4), 2024. This period included two rainfall pulses of 33.6 mm and 64.7 mm, respectively (Fig. 2). For soil sampling, 3-m deep soil samples were collected at 20-cm intervals at each sampling time, using the

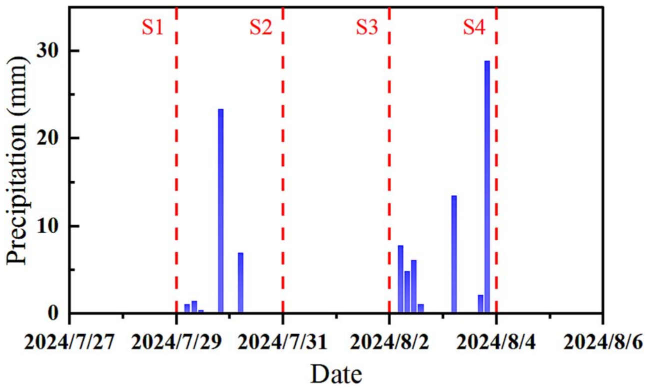

Figure 2 Soil and xylem samples collection schedule.

same approach as for root sampling. A segment of each sample was put into an aluminum container and dried at 105°C for 24 hours until a constant weight was reached, allowing for the determination of soil moisture content (Carter and Gregorich 2007). The leftover soil samples were placed in 100-ml vials, sealed with parafilm, and kept at a low temperature for subsequent water isotope analysis.

Concurrently, on each sampling date, 1-year-old branches were selected from target trees. Once the bark and phloem tissue were removed, the samples were promptly placed into 12-ml glass vials, sealed with Parafilm, and stored at low temperature for the analysis of δ^2H and δ^{18}O water isotopes. Corresponding to soil sampling, three replicate plant xylem samples were collected on each sampling date, resulting in a total of 48 plant xylem samples for water isotope analysis.

Soil and xylem water were extracted via a vacuum condensation system (Orlowski *et al.* 2018). The heating temperatures were set at 205°C for soil samples and 100°C for xylem samples, respectively, with a vacuum pressure below 1 Pa to ensure a water recovery rate between 98% and 102%. The δ^2H and δ^{18}O in soil water were analyzed using a liquid water isotope analyzer (IWA-45EP; Los Gatos Research, Inc., USA). To avoid organic contamination, the δ^2H and δ^{18}O in xylem water were measured by isotope ratio mass spectrometry (253 Plus, Thermo Fisher Scientific Inc., Germany). All results were expressed in Vienna Standard Mean Ocean Water scale, and the measurement precisions for δ^2H and δ^{18}O were ± 1‰ and ± 0.2‰, respectively.

For leaf samples, mature and healthy leaves exposed to sunlight from the target trees were collected in each plot, consistent with the timing of the other sampling activities. Following rinsing with pure water, leaf samples were dried at 70°C for 72 hours until they reached a constant weight. They were then ground into a fine powder for δ^{13}C analysis. The δ^{13}C content in the leaves was measured using an isotope carbon dioxide analyzer (G2131-i, Picarro, Inc., USA) connected to an elemental analyzer (ECS 4024, Costech Analytical Technologies, Italy). The results were expressed relative to the standard sample (Pee Dee Belemnite), with a measurement precision of ±0.1‰.

Data analysis

Degree of coupling coordination model

To assess the interaction between soil water and root system, the coupling coordination degree (D) model was adopted in this study. This model generates a single metric (the coupling coordination D) that quantifies both the strength of interaction (coupling) and the state of synergistic development between two subsystems. FRLD and soil water content (SWC) were selected as the evaluation indices for the root and soil water subsystems, respectively. The D value was calculated using the following formulations (Sun *et al.* 2024):

$$C = \left\{ (f_s * f_r) / \left[(f_s + f_r) / 2 \right]^2 \right\}^{1/2} \tag{1}$$

$$T = \alpha_1 f_s + \alpha_2 f_r \tag{2}$$

$$D = \sqrt{C^* T} \tag{3}$$

In these equations, C is the D of coupling; fs and fr represent the normalized values for the SWC and FRLD, respectively; T is the coordination index; α_1 and α_2 are weighted coefficients. Given the assumed equal importance of soil water and roots for plant function, α_1 and α_2 were both set to 0.5. The value of coupling coordination D ranges from 0 to 1, where a higher D indicates a better coupling relationship. Specifically, D values of 0–0.4 represent unbalanced development, 0.4–0.6 denote transitional development, and 0.6–1.0 indicate balanced development.

Correction for soil and xylem water isotopes

Cryogenic vacuum extraction (CVE) represents the most widely utilized method for isolating water isotopes from soil and xylem tissues (Orlowski *et al.* 2018, Chen *et al.* 2020). However, current studies have shown that this technique introduces systematic offsets in both δ^{18}O and δ^2H values during soil water isotopes sampling (Orlowski *et al.* 2018). These isotopic biases, which correlate with SWC and clay content (CC), can be mathematically quantified as (Wen *et al.* 2021):

$$\Delta\delta^2 H_{soil} = -5.99 - 5.33 \times CC + 21.64 \times SWC \tag{4}$$

$$\Delta\delta^{18} O_{soil} = -0.45 - 0.86 \times CC + 1.86 \times SWC \tag{5}$$

where CC denotes CC (g g^{-1}) and SWC represents gravimetric SWC (g g^{-1}). In contrast, when applied to xylem water extraction, CVE primarily affects δ^2H while exerting negligible influence on δ^{18}O (Chen *et al.* 2020, Wen *et al.* 2022). Given that plant water isotopes reflect the weighted average of their source water, δ^2H offsets can be characterized by (Cai *et al.* 2024, Li 2024):

$$\Delta\delta^2 H_{xylem} = \delta^2 H_{CVE} - a\delta^{18} O_{CVE} - b \tag{6}$$

where parameters a and b represent the slope and intercept of the potential water sources line, respectively. Given that soil water constitutes the sole water source for plants in the sampling area, the potential water sources line is defined by the corrected soil water δ^{18}O and δ^2H values. The final isotopic composition used for analysis is calculated as:

$$\delta = \delta_{CVE} - \Delta\delta \tag{7}$$

Here, δ_{CVE} refers to the isotopic value measured by CVE method, and $\Delta\delta$ represents the computed isotopic offset.

Identification of water sources using CrisPy model

A common approach utilized with the MixSIAR model involves partitioning the continuous soil profile into discrete layers to determine root water uptake sources (Fu *et al.* 2024, Neil *et al.* 2024). However, in the present study, variable infiltration patterns could result in vertically distributed soil water and its isotopic composition lacking distinct stratification characteristics. Misguided layer segmentation could introduce substantial biases and propagate uncertainties in the results. To address this, a continuous isotopic mixing model for root water uptake (CrisPy) was implemented to estimate root water uptake distributions. This framework integrates probability density function (PDF) mixing within a Bayesian inference framework, enabling the direct estimation of continuous root water uptake profiles through solving PDF shape parameters (Fu *et al.* 2024). Advantages include enhanced accuracy and reduced uncertainty compared to traditional layered approaches. Xylem water δ^2H and δ^{18}O values served as mixture input data. For soil water inputs, δ^2H and δ^{18}O values from each sampled soil layer were incorporated into the CrisPy model with non-informative priors. The CrisPy model (Fu and Si 2023) implementation was executed using Python 3.12.

Based on the vertical distribution and temporal dynamics of SWC and water isotopes in the soil profile of the undisturbed area, the soil water in the 0–80 cm layer changes frequently due to the combined effects of precipitation events and evaporation. In contrast, the soil below 80 cm is typically only affected by some large or extreme precipitation events (Wu *et al.* 2022). Therefore, the entire soil profile in the root zone was divided into the shallow layer (0–80 cm) and the deep layer (>80 cm) for analysis. The water uptake proportions of the shallow and deep layers were obtained by integrating the water uptake probability profile derived from the Crispy model over the corresponding depth ranges.

Statistical analysis

A one-way analysis of variance was conducted to examine differences between subsidence and non-subsidence plots using SPSS 25.0 (SPSS Inc., Chicago, IL, USA). Differences with $P < .05$ were considered significant.

Partial least squares structural equation modeling (PLS-SEM) was employed to depict the causal relationships among SWC, FRLD, root-water coupling D, and plant water source under coal mining subsidence. The model structure was hypothesized based on the understanding that subsidence acts as an external disturbance, directly altering SWC and FRLD (Zhang *et al.* 2023). The interaction between SWC and FRLD is synthesized by the coupling coordination D, which in turn is hypothesized to be a direct driver of shifts in plant water source, as root water uptake is mainly controlled by soil water potential and root distribution (Dubbert *et al.* 2023). A feedback path from SWC to FRLD was also included, representing the direct influence of soil water availability on root growth (Fan *et al.* 2017). PLS-SEM analysis was performed using SmartPLS 4.0 software (SmartPLS, Boenningstedt, Germany). The measurement model was evaluated for convergent validity (average variance extracted, AVE > 0.5) and discriminant validity (Fornell-Larker criterion and heterotrait-monotrait ratio). Based on the R^2 values, path coefficients, and significance of indirect effects, the structural model was assessed using bias-corrected confidence intervals.

Results

Soil water content and root distribution

The temporal mean SWC profile of each plot during the sampling period is shown in Fig. 3a and b. In the sandy soil area, the SWC profiles of the subsided and non-subsided plots were similar. The standard deviation of the time-averaged values was larger in the shallow layer and very small in the deep layer (Fig. 3a), indicating that rainfall events primarily caused changes in the SWC of the shallow surface layer. In the loess zone, the non-subsided plots followed a consistent pattern of standard deviation changes from shallow to deep, while the subsided plots exhibited higher standard deviation values at all depths (Fig. 3b), suggesting that the temporal variability of SWC throughout the profile was enhanced.

The average SWC of different soil layers further highlights the differences in soil types. In the sandy soil area, the SWC of the shallow layer (0–80 cm) in the subsided and non-subsided plots was $1.90 \pm 0.30\%$ and $2.25 \pm 0.47\%$, respectively. And the SWC of the deep layer (> 80 cm) in the subsided and non-subsided plots was $1.85 \pm 0.05\%$ and $2.02 \pm 0.05\%$, respectively, with no significant differences (Fig. 3c). In contrast, subsidence did not affect the SWC of the shallow layer in the loess-soil zone, but increased the SWC of the deep layer (> 80 cm) from $5.11 \pm 0.16\%$ to $8.08 \pm 0.97\%$ ($P < .01$) (Fig. 3d).

As depicted in Fig. 4, within both the subsided and non-subsided plots of the sandy soil and loess regions, the FRLD manifested a consistent decline as the soil depth increased. In the sandy soil area, despite a marginal elevation in the FRLD of the subsided plot, it remained largely comparable to the FRLD in the non-subsided plots (Fig. 4a). Conversely, the loess region presented distinct disparities (Fig. 4b). The subsidence condition resulted in notably elevated FRLD values of Mongolian pine in specific soil layers, particularly in the deeper soil strata beneath 80 cm ($P < .05$).

Depth-specific coupling coordination D further elucidates the subsidence-induced modifications in root-water coordination dynamics (Fig. 5). In sandy soil, the non-subsided profiles exhibited consistent unbalanced development ($D < 0.4$) across the entire depth range (Fig. 5a). However, subsidence shifted most D values in the deep soil layers (80–160 cm) to the transitional range ($0.4 \le D < 0.6$), indicating improved system structure. In the loess area, non-subsided plots maintained transitional coordination ($D > 0.4$) only in the shallow layers (0–80 cm), while subsidence extended this coordinated zone to a depth of 140 cm (Fig. 5b). In summary, the scope of unbalanced development zones in the soil profiles of subsided plots (both sandy soil and loess) was reduced, demonstrating that subsidence optimized the root-water synergy in deep soil layers.

Isotopic composition of xylem and soil water

Figure 6 illustrates the associations between δ^2H and δ^{18}O across various water bodies, compared with the Global Meteoric Water Line (GMWL: $\delta^2\text{H} = 8\delta^{18}\text{O} + 10$) and the Local Meteoric Water Line (LMWL: $\delta^2\text{H} = 7.67\delta^{18}\text{O} + 5.91$ (Zhao *et al.* 2022)). In sandy soil areas, the Soil Water Lines (SWLs) for subsidence and non-subsidence plots deviated from the GMWL and LMWL, indicating that soil water within the root zone has undergone intense evaporation. Similarly, the SWL of non-subsidence plots in the loess area also deviated from the GMWL and LMWL. However, the SWL of subsidence plots in the loess area was very close to the GMWL and LMWL, and water at different soil layers distributed along the meteoric water lines. This indicates that soil water in the entire root zone is directly replenished by precipitation and rarely affected by evaporation, demonstrating the characteristics of preferential flow recharge.

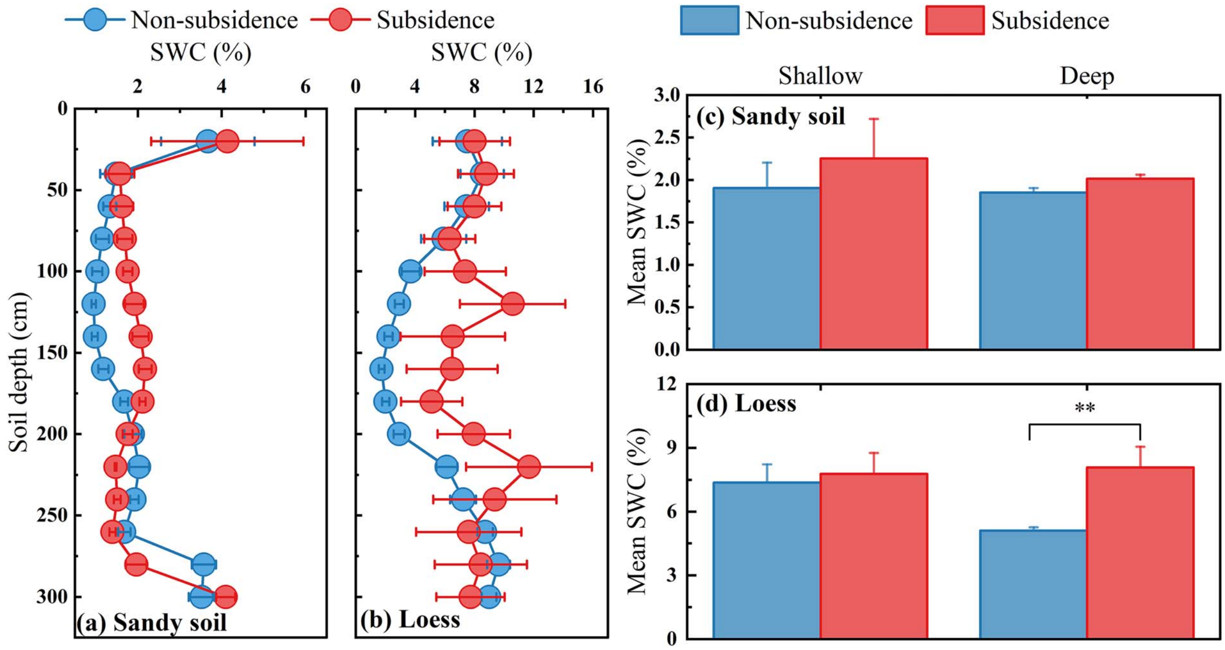

Figure 3 Gravimetric SWC under subsidence and non-subsidence plots in sandy soil and loess zones. Error bars denote the standard deviation derived from the mean.

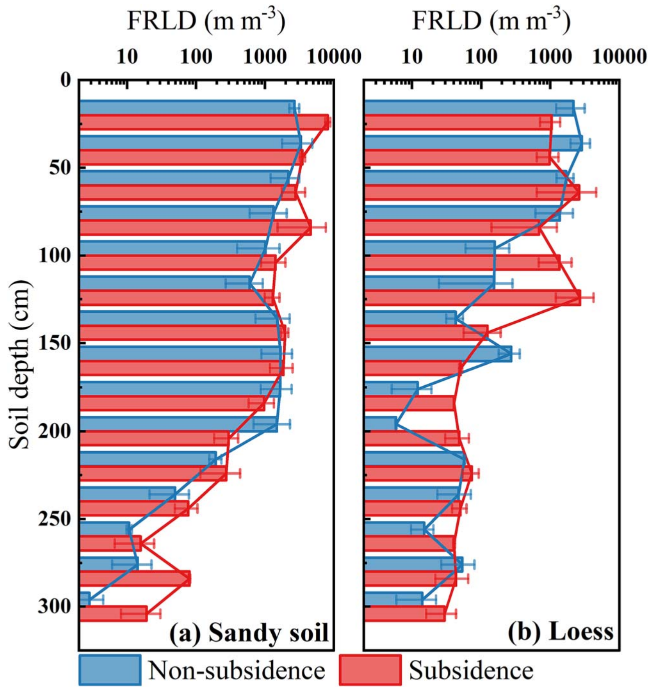

Figure 4 Vertical variation in FRLD across subsidence and non-subsidence areas in sandy soil and loess regions. Error bars denote the standard deviation derived from the mean.

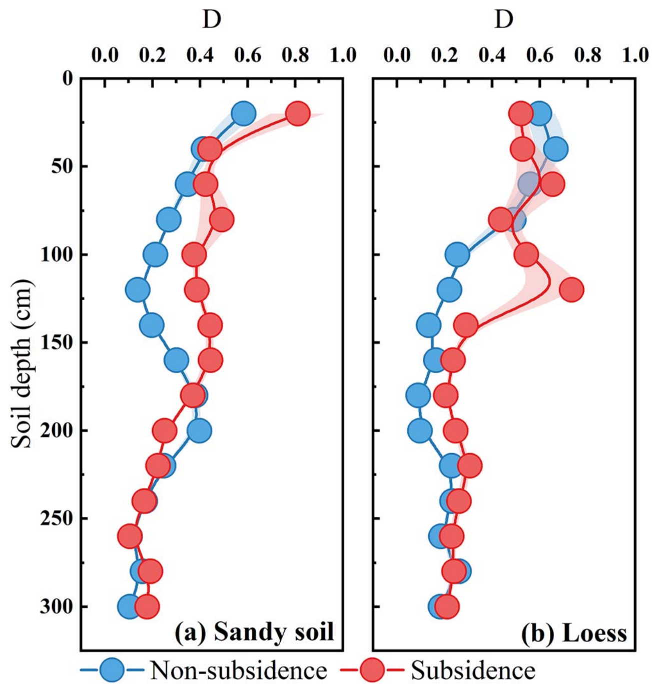

Figure 5 Vertical distribution of coupling coordination D of soil water and fine root length under different plots. Error bars represent the standard error of D.

Variations in the soil profile also emphasize water migration characteristics consistent with the aforementioned SWC patterns, as influenced by soil types under the impact of subsidence (Fig. 7). In the sandy soil area, the profiles of the isotopic compositions of soil water in the subsidence plot and the non- subsidence plot exhibited comparable temporal and spatial trends (Fig. 7a). Temporally, the isotopic composition of shallow soil water (0–80 cm) varied greatly, indicating the interference of precipitation recharge and evaporation dynamics, while the deep soil water (> 80 cm) isotopic composition hardly changed over time. Spatially, the δ^2H and δ^{18}O values of soil water in both plots became more depleted as depth increases. Likewise, subsidence had little impact on the isotopic composition of Mongolian pine xylem water in sandy soil areas, which remained close to that of deep soil water. However, in the loess area, the temporal and spatial

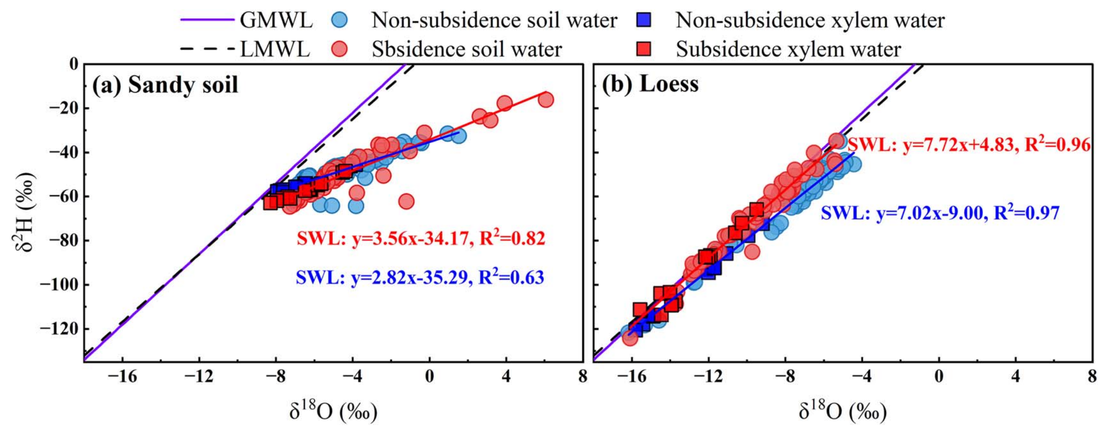

Figure 6 Relationships between δ^2H and δ^{18}O in different water bodies, compared with the global meteoric water line (GMWL: $\delta^2H = 8\delta^{18}O + 10$) and the LMWL ($\delta^2H = 7.67\delta^{18}O + 5.91$ (Zhao *et al.* 2022)) under subsidence and non-subsidence plots in sandy soil and loess zones.

variations in the isotopic composition profiles of soil water differed entirely between subsidence and non-subsidence plots (Fig. 7b). For the non- subsidence plot, the soil water δ^2H and δ^{18}O still only had obvious temporal variations in the shallow soil, while the isotopic composition of soil water in the subsided plot showed drastic temporal fluctuations throughout the profile, indicating that the subsidence allows rainfall to penetrate the entire profile. Meanwhile, deep soil water in the subsidence plot exhibited a more depleted isotopic composition. While the isotopic composition of Mongolian pine xylem water showed no significant difference between subsidence and non-subsidence plots, the xylem water isotopes in the subsidence plot overlapped more extensively with those of deep soil water.

Water uptake proportion of Mongolian pine

Water sources of Mongolian pine estimated by the CrisPy model showed that the impact of subsidence on the uptake proportion from potential soil layers varied depending on the soil type (Fig. 8). In the sandy soil area, the uptake proportions from shallow soil (0–80 cm) and deep soil (> 80 cm) in the non-subsidence plot were 20.58 ± 4.70% and 79.43 ± 3.83%, respectively (Fig. 8a). For Mongolian pine in the subsidence plot, shallow and deep soil contributed 17.31 ± 3.13% and 82.69 ± 1.52% to its water uptake, respectively, with the proportion of water uptake from deep soil in the subsidence plot showing a slight increase compared to that in the non-subsidence plot. Conversely, in the loess area, the uptake proportion from deep soil in the subsided plot is significantly greater than that in the non-subsided plot, which are 26.36 ± 1.98% and 16.23 ± 1.91%, respectively ($P < .01$) (Fig. 8b).

PLS-SEM analysis further clarified the contributions of various factors and revealed that the D of disturbance caused by coal mining subsidence is influenced by soil type (Fig. 9). In the sandy soil area, coal mining subsidence had no direct impact on SWC or FRLD, and there was no pathway through which mining subsidence affects soil water and root distribution, thereby influencing their coupling D and water uptake sources (Fig. 9a). However, as shown in Fig. 9b, in the loess soil area, coal mining subsidence exerted a significant positive effect on SWC, with a standardized coefficient of 0.906 ($P < .01$). With the increase in SWC, FRLD increased significantly, with a standardized coefficient of 0.337 ($P < .05$). The combined increase in SWC and FRLD promoted the enhancement of their coupling coordination D, with standardized coefficients of 0.220 and 0.859 respectively ($P < .01$), and ultimately led to an increase in root water uptake (standardized coefficient = 0.738, $P < .01$).

δ^{13}C in plant leaves

The δ^{13}C value of Mongolian pine leaves in the subsided plot of the sandy soil region was −26.11 ± 0.22‰, which was significantly lower than that in the non-subsided plot (−25.64 ± 0.14‰) ($P < .05$) (Fig. 10). Similarly, in the loess area, subsidence significantly reduced the δ^{13}C value of Mongolian pine leaves ($P < .01$), with a more substantial decrease, as the values in the subsided and non-subsided plots were – 26.12 ± 0.32‰ and – 24.56 ± 0.31‰, respectively (Fig. 10).

Discussion

Dynamics of soil water response to subsidence across different soil types

Our observations demonstrate that land subsidence notably elevates water in deep soil horizons (> 80 cm) (Fig. 3) and expands the infiltration scope of precipitation, particularly within loess regions (Fig. 6 and Fig. 7). This finding aligns with prior research emphasizing the role of subsidence in augmenting deep soil water retention, primarily through alterations in vadose zone architecture induced by underground mining activities (Zhang *et al.* 2022). The formation of cracks and fissures in the soil matrix, a direct consequence of subsidence, can establish preferential flow pathways that facilitate the downward movement of water into deeper soil layers (Xiang *et al.* 2019). The stronger response in loess areas results from differences in soil hydrology. Sandy soils, characterized by poor aggregation and coarse texture, exhibit high permeability and low water-holding capacity, which promote rapid water translocation (Dong *et al.* 2022). Their weak particle cohesion limits subsidence-induced pore structural modifications, thereby constraining SWC variations. In contrast, loess with higher silt fractions and superior water retention properties undergoes significant pore restructuring during subsidence due to their inherent structural stability (Zhao *et al.* 2016, Tong *et al.* 2024). This process generates novel soil pores that amplify deep soil moisture dynamics. These findings highlight the pivotal function of soil characteristics in regulating the effects of coal mining subsidence on soil water regimes.

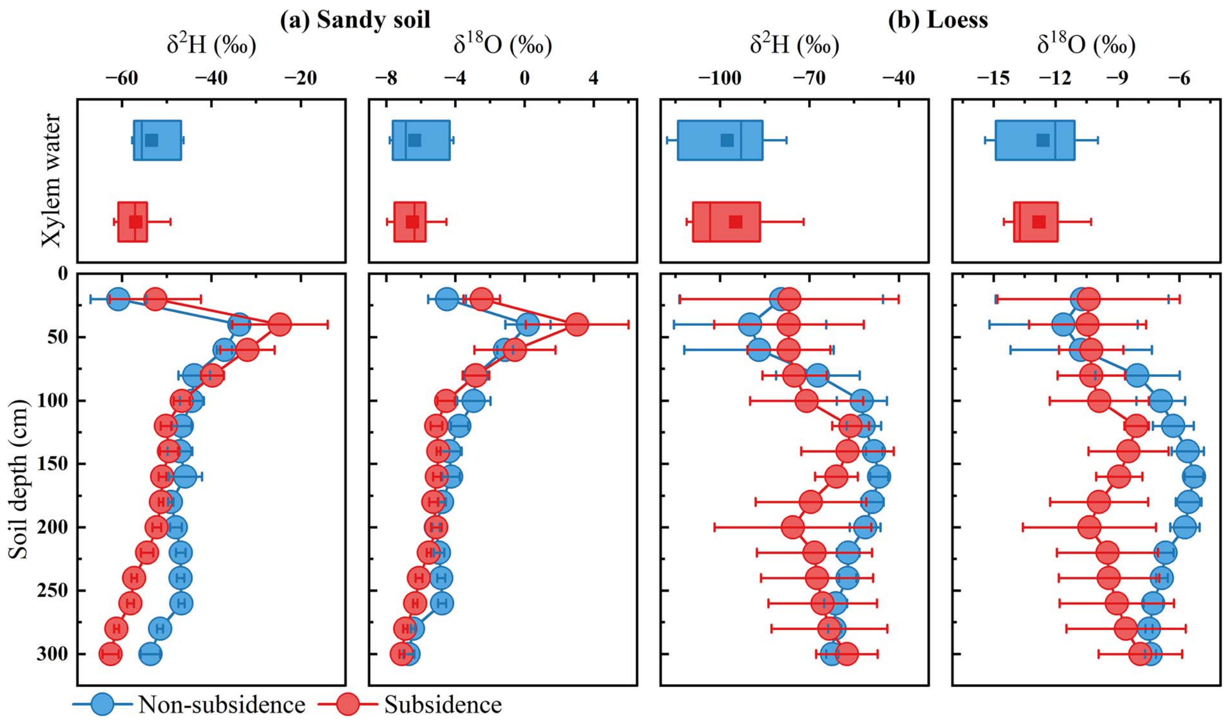

Figure 7 δ^2H and $^\delta$18O for soil profile and plants xylem water under subsidence and non-subsidence plots in sandy soil and loess zones. Error bars denote the standard deviation derived from the mean.

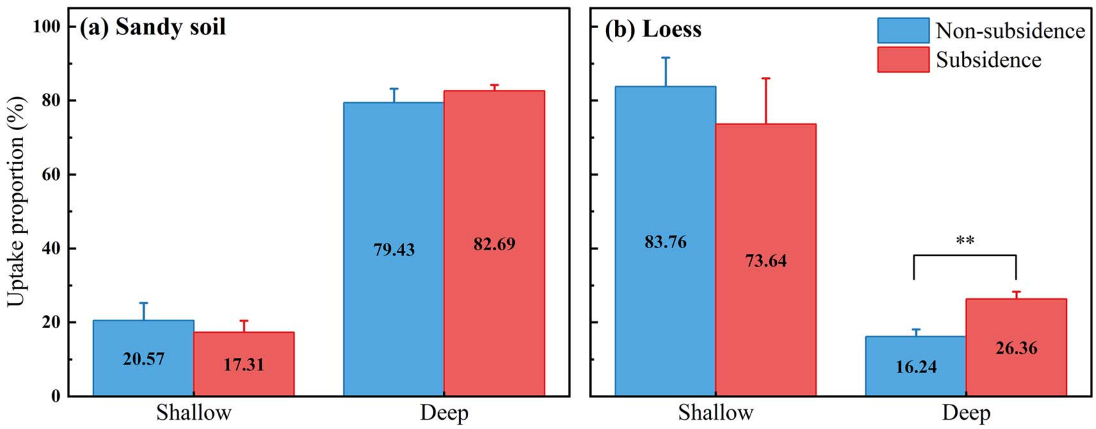

Figure 8 Uptake proportion from different soil layers under subsidence and non-subsidence plots in sandy soil and loess zones. Error bars denote the standard deviation derived from the mean.

Effect of subsidence on Mongolian pine root systems across different soil types

In this study, fine root length density (FRLD) showed comparable values between non-subsidence and subsidence plots in sandy soil region. However, in loess region, subsidence plots exhibited significantly higher FRLD in deep soil layers compared to non-subsided areas (Fig. 4). The observed soil-type dependent deep root proliferation likely results from multiple interacting factors. Enhanced deep soil water availability in subsided loess zones creates favorable conditions for root extension into deeper horizons (Chen *et al.* 2023). Concurrently, subsidence-induced vadose zone structural modifications increase soil porosity and aeration, providing physical space for root system expansion (Pierret *et al.* 2016). The formation of stable cracks and fissures in subsidence-affected areas may facilitate root penetration by reducing mechanical resistance and improving access to deeper soil layers (Lu *et al.* 2019). Collectively, these findings demonstrate that coal mining subsidence alters root growth environments, particularly in loess regions, by promoting deeper root systems that enhance plants' potential to access these water resources.

Water use strategies in the Mongolian pine response to subsidence across different soil types

Tree water use strategies can be used to assess their environmental adaptability and inform ecological restoration initiatives. In general, coal mining subsidence increased the proportion of deep soil water absorption in Mongolian pines, especially in loess areas (Fig. 8). This observed shift was validated and mechanistically elucidated by our SEM analysis (Fig. 9), which revealed contrasting causal pathways

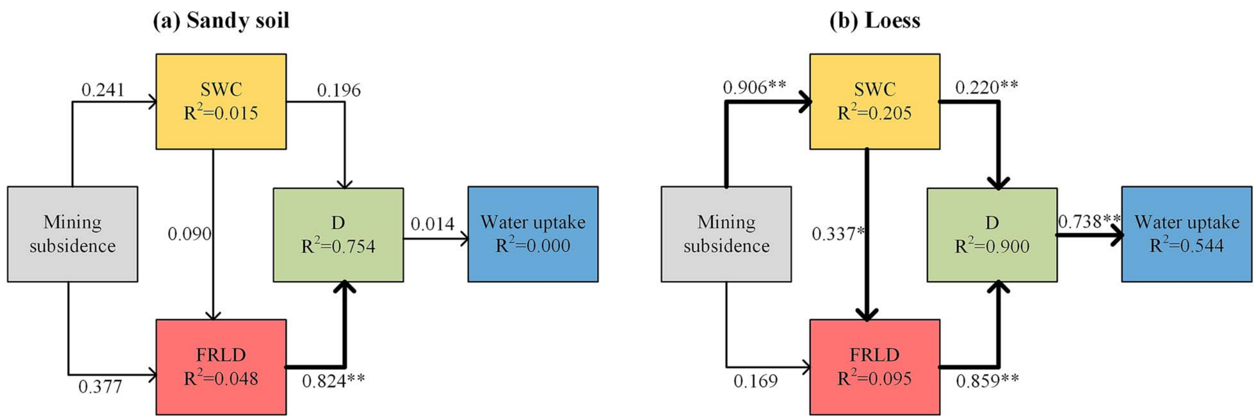

Figure 9 SEM path diagram of plant water use caused by mining subsidence in in sandy soil (a) and loess (b) zones. Numbers represent standardized path coefficients, significance values are indicated by * ($P < .05$), **($P < .01$). SWC: Soil water content; FRLD: Fine root length density; D: Coupling coordination D of SWC and FRLD.

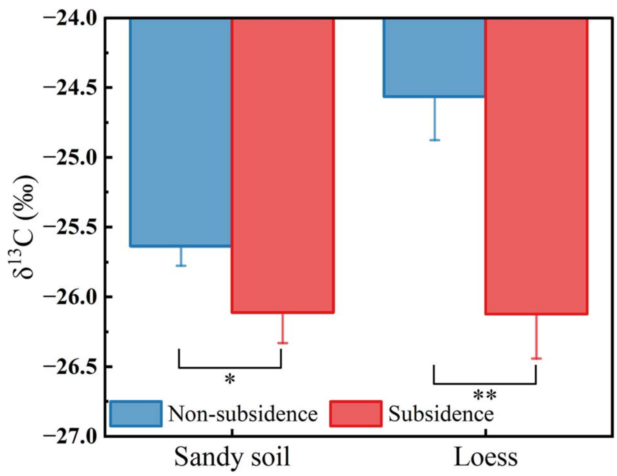

Figure 10 Variations in leaf δ^{13}C under subsidence and non-subsidence plots in sandy soil and loess zones. Error bars denote the standard deviation derived from the mean.

between soil types. In sandy soil, no significant path linked subsidence to changes in SWC, FRLD, or water uptake, confirming a buffered response. In loess soil, a clear positive cascade was identified: subsidence directly increased SWC, which promoted FRLD. The concurrent rise in both SWC and FRLD synergistically enhanced their coupling coordination D, which in turn directly and strongly drove the increased reliance on deep soil water. Thus, the model substantiates a cause-effect chain where subsidence alters the vadose zone, and the strengthened soil-root synergy (quantified by D) is the key mechanism enabling the shift in the main water source. Studies on the water uptake from differing sources of Mongolian pine in Horqin Sandy Land similar to those in this study had also shown that the proportion of water uptake sources changes with variations in soil water conditions (Song *et al.* 2016, 2020). Meanwhile, other research revealed that the spatial distribution of the root system of Mongolian pine exerts a significant influence on its water uptake sources (Zhang *et al.* 2021). Plant water uptake represents an integration of physical and biological processes (Couvreur *et al.* 2018, Agee *et al.* 2021). Roots act as intermediaries for water absorption, with their vertical distribution directly influencing water source utilization (Gale and Grigal 2011, Heinen 2014). Soil water availability reflects the soil's capacity to supply water, where low water content imposes limitations on plant water uptake (Demir *et al.* 2024, Sperling *et al.* 2025). Thus, Mongolian pine in loess regions showed more distinct changes in deep soil water use and root traits in response to subsidence, leading to increased uptake of deep soil water.

Leaf δ^{13}C values mirror plants' long-term water use responses to environmental conditions and act as markers of water use efficiency. (Kenney *et al.* 2014, Davidson *et al.* 2023). In arid regions, increasing leaf δ^{13}C values correlated strongly with water stress, indicating stomatal limitations (Wu *et al.* 2022). In both soil types, subsidence plots showed significantly lower leaf δ^{13}C values in Mongolian pine compared to non-subsidence plots (Fig. 10), suggesting reduced water stress. Furthermore, subsidence effects on leaf δ^{13}C values varied with soil type. Loess regions displayed more dramatic δ^{13}C reductions compared to sandy soil regions (Fig. 10). This discrepancy aligned with deep soil water uptake proportions, indicating that subsidence-affected Mongolian pine mitigates water stress and maintains stomatal openness by accessing more deep soil water. Extensive research confirmed that increased deep soil water utilization helps plants alleviate drought stress and sustain stem water potential and photosynthetic rates (Shao *et al.* 2023, Yang *et al.* 2023b). In brief, compared to sandy regions, Mongolian pine in loess regions demonstrated more pronounced deep soil water utilization and greater water stress alleviation under mining subsidence. This soil-type dependent response highlights the necessity for distinct vegetation maintenance or restoration strategies in subsided areas across different soil matrices.

Limitations and future perspectives

This study establishes a soil-dependent mechanistic framework in which mining subsidence enhances deep soil water availability, promotes root proliferation, and strengthens their synergistic coupling, ultimately leading to increased plant reliance on deep soil water (Fig. 9). Given the focused nature of this case study, the conclusions are derived from a relatively small dataset with limited replication within a single, albeit geologically distinct, mining area. The main limitations arise from the focused sampling design and the specific environmental context. By concentrating on areas immediately adjacent to subsidence fractures at two sites, this work effectively characterizes the direct effects of pronounced vadose zone alteration. Consequently, the findings are most representative of a high-disturbance scenario (proximal to fractures) and do not capture the full gradient of subsidence intensity. The ecohydrological benefits are likely to diminish with increasing distance from fractures, a spatial gradient not quantified here. Moreover, as a snapshot study, it does not address how these effects may evolve over time as fractures weather or seal. Furthermore, the observed outcomes and the strength of the quantified relationships may be context-dependent. For instance, the long-term net effect of enhanced infiltration on deep soil water storage is contingent on the regional balance between precipitation and evapotranspiration (Chen *et al.* 2025). Similarly, site-specific factors such as land-use history, which can profoundly alter soil structure, organic matter, and pre-existing subsurface heterogeneity (Landgraf *et al.* 2023), may modulate how the soil–plant system responds to subsidence.

These defined boundaries highlight clear priorities for future research. To move from a focused case study toward a generalizable understanding, subsequent work should: (i) quantify the spatial gradient of impacts by sampling along transects at varying distances from fractures; (ii) initiate long-term monitoring to understand the temporal dynamics of the system; and (iii) test the robustness of the proposed mechanistic framework across a wider range of climatic regimes and land-use histories. Such efforts are crucial for developing predictive models and designing effective, site-specific restoration strategies for mining-affected landscapes.

Conclusion

Using stable isotopes (δ^2H, δ^{18}O, and δ^{13}C) combined with SWC and root distribution, this study investigated differences in water use strategies of Mongolian pine in coal mining subsidence and non-subsidence plots across sandy soil and loess regions. We assume that coal mining subsidence altered vadose zone structures which led to observed increased water availability in deep soil layers, thereby promoting root growth and strengthening the coupling between the two, especially in loess areas. Such shifts resulted in a marked rise in the uptake of deep soil water by Mongolian pine in loess subsided zones, while only a slight rise occurred in sandy subsided zones. Significantly depleted leaf δ^{13}C values in subsided areas indicated reduced water stress due to the utilization of deep soil water, particularly in loess regions. Coal mining subsidence exerted beneficial effects on deep soil water accessibility and plant water utilization, especially in water-scarce arid regions. Meanwhile, soil-type dependent response highlights the necessity for distinct vegetation maintenance or restoration strategies in subsided areas across different soil matrices. These findings improve understanding of plant survival strategies and their relationships with water resources in subsided areas, offering useful insights for sustainable land and water management in mining-affected regions.

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Conflict of interest

None declared.

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Data availability

Data will be made available on request.

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